

B MMLU

		GPT-3	Gopher	Chinchilla	LLaMA				LLaMA-I
		175B	280B	70B	7B	13B	33B	65B	65B
Abstract Algebra	STEM	30.0	25.0	31.0	29.0	34.0	32.0	34.0	31.0
Anatomy	STEM	48.0	56.3	70.4	37.0	45.9	51.9	57.8	62.2
Astronomy	STEM	49.0	65.8	73.0	33.6	46.1	61.8	72.4	81.6
Business Ethics	Other	46.0	70.0	72.0	40.0	45.0	56.0	57.0	72.0
Clinical Knowledge	Other	48.0	67.2	75.1	35.1	45.7	57.4	65.3	69.1
College Biology	STEM	45.0	70.8	79.9	37.5	45.1	58.3	68.8	81.9
College Chemistry	STEM	26.0	45.0	51.0	32.0	30.0	45.0	50.0	45.0
College Computer Science	STEM	46.0	49.0	51.0	29.0	39.0	45.0	47.0	51.0
College Mathematics	STEM	34.5	37.0	32.0	33.0	32.0	40.0	35.0	36.0
College Medicine	Other	48.0	60.1	66.5	30.6	42.8	52.0	54.3	63.0
College Physics	STEM	28.0	34.3	46.1	26.5	18.6	28.4	36.3	46.1
Computer Security	STEM	57.0	65.0	76.0	45.0	65.0	66.0	79.0	79.0
Conceptual Physics	STEM	36.5	49.4	67.2	36.6	41.3	51.5	59.6	66.4
Econometrics	Social Science	33.0	43.0	38.6	23.7	27.2	35.1	40.4	52.6
Electrical Engineering	STEM	50.0	60.0	62.1	26.9	40.7	49.7	53.8	60.7
Elementary Mathematics	STEM	30.0	33.6	41.5	24.3	24.9	36.0	37.8	42.9
Formal Logic	Humanities	29.0	35.7	33.3	27.0	33.3	34.1	44.4	47.6
Global Facts	Other	37.0	38.0	39.0	29.0	35.0	35.0	39.0	40.0
High School Biology	STEM	48.0	71.3	80.3	34.5	52.6	67.7	73.9	82.9
High School Chemistry	STEM	33.0	47.8	58.1	28.1	28.6	41.9	40.4	44.8
High School Computer Science	STEM	39.0	54.0	58.0	31.0	48.0	60.0	67.0	73.0
High School European History	Humanities	54.0	72.1	78.8	44.2	61.8	73.9	78.8	86.1
High School Geography	Social Science	58.0	76.8	86.4	34.3	54.6	70.7	77.8	87.9
High School Government And Politics	Social Science	58.0	83.9	91.2	44.6	66.3	82.9	88.1	92.8
High School Macroeconomics	Social Science	40.5	65.1	70.5	35.4	44.4	56.9	65.9	69.2
High School Mathematics	STEM	28.0	23.7	31.9	24.8	23.7	27.0	34.4	37.0
High School Microeconomics	Social Science	42.0	66.4	77.7	31.9	47.5	55.5	68.9	78.6
High School Physics	STEM	28.0	33.8	36.4	26.5	28.5	35.8	37.1	41.7
High School Psychology	Social Science	61.0	81.8	86.6	47.3	60.9	76.2	82.2	87.9
High School Statistics	STEM	30.5	50.0	58.8	35.2	30.1	45.4	58.3	59.3
High School Us History	Humanities	53.0	78.9	83.3	39.7	58.3	77.9	83.8	90.7
High School World History	Humanities	56.0	75.1	85.2	40.9	66.2	79.3	83.1	89.0
Human Aging	Other	50.0	66.4	77.6	40.8	54.7	67.7	69.5	72.2
Human Sexuality	Social Science	54.0	67.2	86.3	36.6	58.8	64.1	77.9	87.0
International Law	Humanities	55.5	77.7	90.9	51.2	62.8	72.7	79.3	87.6
Jurisprudence	Humanities	55.0	71.3	79.6	38.9	51.9	70.4	73.2	85.2
Logical Fallacies	Humanities	48.0	72.4	80.4	39.3	52.8	68.1	77.3	80.4
Machine Learning	STEM	31.0	41.1	41.1	23.2	31.3	39.3	49.1	52.7
Management	Other	56.0	77.7	82.5	35.0	66.0	77.7	82.5	83.5
Marketing	Other	60.0	83.3	89.7	46.6	71.8	83.3	85.9	92.7
Medical Genetics	Other	40.0	69.0	69.0	43.0	52.0	67.0	67.0	68.0
Miscellaneous	Other	60.0	75.7	84.5	42.4	65.4	78.5	82.1	84.3
Moral Disputes	Humanities	44.5	66.8	77.5	40.2	50.9	66.2	72.3	76.9
Moral Scenarios	Humanities	26.0	40.2	36.5	24.3	30.1	38.2	48.9	55.9
Nutrition	Other	47.0	69.9	77.1	37.6	51.6	62.8	67.3	74.5
Philosophy	Humanities	51.0	68.8	79.4	39.9	54.0	66.2	74.0	79.1
Prehistory	Humanities	53.0	67.6	81.2	36.1	51.5	67.0	75.3	79.0
Professional Accounting	Other	33.0	44.3	52.1	25.9	35.8	43.6	46.5	56.0
Professional Law	Humanities	34.5	44.5	56.5	30.2	38.0	45.9	49.1	54.4
Professional Medicine	Other	36.0	64.0	75.4	44.5	50.4	54.0	61.4	70.6
Professional Psychology	Social Science	44.5	68.1	75.7	35.1	47.7	62.9	65.7	71.4
Public Relations	Social Science	48.0	71.8	73.6	40.9	60.9	67.3	73.6	74.6
Security Studies	Social Science	52.0	64.9	75.9	31.8	53.9	65.3	71.8	77.6
Sociology	Social Science	53.0	84.1	91.0	46.8	61.2	78.6	78.6	88.1
Us Foreign Policy	Social Science	69.0	81.0	92.0	46.0	80.0	83.0	86.0	87.0
Virology	Other	46.0	47.0	53.6	30.1	43.4	50.0	53.0	57.8
World Religions	Humanities	55.0	84.2	87.7	50.9	67.8	81.3	81.3	84.2
Humanities		40.6	56.2	63.6	34.0	45.0	55.8	61.8	67.4
STEM		36.7	47.4	54.9	30.5	35.8	46.0	51.7	56.6
Social Science		50.5	71.9	79.3	38.3	53.8	66.7	72.9	79.2
Others		49.0	66.1	73.9	38.1	53.3	63.4	67.4	72.6
All		43.9	60.0	67.6	35.1	46.9	57.8	63.4	68.9

Table 16: MMLU. Detailed 5-shot results per domain on the test sets.